MILITARY SPECIFICATION

SEALS, INTERTUBE FOR LAUNCHER TUBE MARK 4 MODS 0, 1, 2, AND 3, AND MARK 8 MODS 0 AND 1

This amendment forms a part of Military specification MIL-S-21520C (Wep), 7 June 1962, and has been concurred in by the Bureau of Naval Weapons.

Page 3, paragraph 2.2: Delete:

"NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing office, Washington 25, D. C.

Page 4, paragraph 3.2.1: Delete and substitute:

"3.2.1 Preproduction Sample - Unless otherwise specitied in the contract or order, preproduction samples of the upper and lower seals shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification (refer to 4.2.1)."

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Page 5, paragraph 3.2.7: Delete.

Page 5, add new paragraph 3.2.11:

"3.2.11 Seal Test Fixture - Unless otherwise specified, any fixture used in the verification of the requirements of this specification shall support and contain the seal in exactly the same manner as a seal installed in the tactical missile launcher tube."

Page 5; paragraph 3.3.1: Delete and substitute:

"3.3.1 Leakage - The intertube seals shall withstand a hydrostatic pressure of 195 to 205 psig for 10 minutes without Mileage OP structural damage."

Page 6, paragraph 3.3.2.1: Delete and substitute:

"3.3.2.1 Steady Pressure - The intertube seals shall withstand a hydrostatic pressure of 195 to 205 psig applied continuously for 100 hours without leakage or structural damage. Thereafter, the seals shall be capable of meeting all performance requirements of this specification, except that the 100 hour steady pressure requirement shall not be in addition to the requirement specified in 3.3.2.3."

Page 6 paraagraph 3.3.2.2: Delete and substitute:

"3.3.2.2 Rapidly Applied Pressure - The intertube seals shall withstand 25 cycles of rapidly applied hydrostatic pressure at a rate of 1 cycle per 30 minutes. Thereafter, the seals shall show no leakage or structural damage and shall be capable of meeting all performance requirements or this specification. A cycle of rapidly applied Hydrostatic pressure shall consist of raising the pressure to 30 psig and holding this pressure for 1 minute, then increasing the pressure to 130 ± 5 psig within 0.5 to 0.7 second and holding this pressure for 15 seconds, then decreasing the pressure to 0 psig within 2 minutes.

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Page 6, paragraph 3.3.2.3: Delete and substitute:

“3.3.2.3 Pressure Cycling - The intertube seals shall withstand 50 cycles of hydrostatic pressure. Thereafter the seals shall show no leakage or structural damage and shall be capable of meeting all performance requirements of this specification, except that the 50 pressure cycles need not be in addition to the requirement specified in 3.3.2.1. A pressure cycle shall consist of increasing the pressure from 0 psig to between 195 and 205 psig and holding this pressure for 2 hours, then decreasing the pressure to 0 psig and holding 0 psig for 30 minutes before the next cycle is started."

page 6, paragraph 3.3.3: Delete and substitute:

“3.3.3 Flexure - The Intertube seals shall withstand 300 cycles of flexure at the rate of 1 cycle per minute. Thereafter, the seals shall show no structural damage and shall be capable of meeting all performance requirements of this specification. A cyyle of flexure shall consist of moving the inner circumference of the seals relative to the periphery, through an 8 by 12-inch elliptical path with the major axis of the ellipse perpendicular to the diameter of the seal."

Page 7, paragraph 3.4.6: Delete and substitute:

“3.4.6 Heat and Pressure Cycle - The lower intertube seal shall withstand the environment of 5 consecutive steam heat/pressure cycles without damage. Thereafter, the seeal shall be capable of meeting all performance requirements of this specification. A steam heat/pressure cycle shall consist of a change in pressure from atmospheric to 115 ± 15 psig and return to atmospheric coincident with a temperature change from ambient to +325 ± 25 degrees Fahrenheit to +210 ± 20 degrees Fahrenheit within a maximum of 1 second and then to amibient temperature within a maximum of 10 minutes.

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Page 8, paragraph 4.2 line 11:

Following "NAVWEPS OP 400" insert "except as modified herein."

Page 9, paragraph 4.3.1: Delete and substitute:

“4.3.1 Production Tests - These tests are to be accomplished on each upper and lower seal being submitted for acceptance under the contract. Production tests shall be either performed by the contractor and witnessed by the Government Inspector, or these tests may be conducted by the Government Inspector. These tests are detailed in 4.5, An assembly rejected because of superficial damage, such as pin holes evidenced by minor leakage, that may develop during these tests may be reworked and resubmitted for acceptance test, but shall not receive more than one such action. Any structural damage to the seal that may develop during production testing shall result in rejection of the production lot with no recourse for resubmission and shall cause immediate cessation of’ production until determination of cause of failure is made and appropriate correction is approved by the procuring activity."

Page 9, paragraph 4.3.1.1: Delete and substitute:

“4.3.1.1 Periodic Production Tests - Periodic production tests are those which are accomplished on samples selected as specified in 4.2.3, which are representative of the production of the seals, to determine that the production meets the requirements of this specification. These tests are detailed in 4.5 and 4.7. Superficial damage, such as pin holes evidenced by minor leakage, that may develop during the tests of the requirements specified in 3.3.2.3, may be

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repaired if necessary to meet leakage requirements. Any structural damage to the sample shall result in rejection of the production lot with no recourse for resubmission and shall cause immediate cessation of production until determination of cause of failure is made and appropriate correction is approved by the procuring activity."

page 13, paragraph 4.8: Delete.

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